VINCENT DAVIDSON

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RESEARCH PROFILE

Ph.D. candidate in Computer Science and Engineering focused on applied AI/ML research within enterprise data and analytics environments. Research interests center on trustworthy AI, context engineering, knowledge graphs, information retrieval, human-AI teaming, explainable machine learning, and responsible GenAI adoption. At Nationwide's Enterprise Analytics Office and Enterprise Data Office, translates research questions into proofs of concept, governance patterns, and scalable platform capabilities that improve the relevance, reliability, cost awareness, and responsible use of AI.

RESEARCH AND IMPACT

- **Trusted data and AI research —** Developed a research-driven Trusted Data Environment proof of concept for trusted data and AI-assisted context retrieval; received Nationwide's Inaugural Trusted Data Impact Spotlight Award through the 2026 Enterprise Hackathon.
- **Context engineering —** Investigate token heterogeneity, token-per-minute limits, context retrieval cost, and enterprise RAG behavior to inform practical strategies for scalable, cost-aware, and responsible AI adoption.
- **AI/ML research translation —** Research and prototype how knowledge representation, information retrieval, and governed data reasoning can improve enterprise AI relevance, interoperability, explainability, and trust.
- **Responsible AI enablement —** Advance applied AI governance through experimentation and operating guidance for Databricks Genie, dashboard and Genie space lifecycle management, LLM restrictions, workspace rate limits, token quotas, and platform usage patterns.
- **Research-to-production —** Connect research prototypes to analytic lifecycle concerns including data readiness, model governance, monitoring, adoption, and downstream consumption across EAO and EDO environments.

EDUCATION

The Ohio State University — Columbus, OH August 2024 – Current
Ph.D., Computer Science — Ph.D. Candidate
College of Engineering, Department of Computer Science | Advisor: Dr. Christopher Stewart

Clemson University — Clemson, SC August 2022 – August 2024
M.S., Computer Science

Winston-Salem State University — Winston-Salem, NC August 2012 – May 2018
B.S., Computer Science

RESEARCH AREAS AND METHODS

Trustworthy and responsible AI; context engineering; information retrieval; knowledge graphs and semantic reasoning; RAG and conversational AI; explainable and interpretable machine learning; human-AI teaming; NLP and semantic parsing; model evaluation; AI governance; requirements engineering; literature review and research synthesis; proof-of-concept design; stakeholder-centered technology evaluation.

PROFESSIONAL EXPERIENCE

The Ohio State University — Columbus, OH August 2026 - Current
Graduate Research Associate — Translational Data Analytics Institute (TDAI)

- Developed and evaluated Python and R code, exploratory data analysis, data dictionaries, data-preparation workflows, sprint plans, retrospectives, written reports, executive summaries, technical documentation, sponsor presentations, modeling artifacts, deployment plans, final code, and final reports for rigor, reproducibility, maintainability, and completeness.
- Held office hours and mid-sprint check-ins, monitored blockers and team progress, supported pitch, mid-semester review, and final presentation sessions, and served as an escalation point for team conflicts and sponsor concerns; provided feedback within course policy timelines.
- Supported TDAI faculty in applied research through literature reviews, data collection, analysis, visualization, reporting, grant proposal preparation, and scholarly outputs connected to translational analytics and Capstone activities.
- Contributed to course development by creating modules, assignments, rubrics, instructional materials, reusable AI workflow templates, starter repositories, and technical guides; assisted faculty with integrating emerging AI tools into the curriculum.

Nationwide — Columbus, OH February 2026 – Current
Specialist, Technology Engineer — Enterprise Data Office

- Conduct applied research and research-driven experimentation on Databricks AI capabilities, enterprise RAG architectures, foundation-model use, secure LLM deployment, governed data reasoning, and GenAI orchestration.
- Translate research findings into reusable guidance and guardrails that help teams adopt AI capabilities with greater consistency, lower operational risk, stronger governance, and improved cost awareness.
- Partner with stakeholders to frame research questions and business opportunities, identify data and platform requirements, evaluate solution options, and connect the right technical specialists to move work forward.

- Design and optimize ML/GenAI workflows using MLflow tracking, model lifecycle management, AutoML, distributed training, model serving, monitoring, and Feature Store governance.
- Use context-engineering principles to evaluate how source selection, retrieval quality, knowledge representation, and context cost affect enterprise AI usefulness and trust.
- Establish repeatable lifecycle governance for Databricks dashboards and Genie spaces through inventory, activity tracking, ownership clarification, and remediation recommendations across development, test, and production environments.

Nationwide — Columbus, OH December 2024 – February 2026
Senior Analyst, Technology Engineer — Enterprise Data Office

- Supported applied research and platform experimentation focused on Databricks AI/ML adoption, enterprise usage patterns, reliability, observability, and operational readiness.
- Helped implement LLM restrictions, workspace rate limits, token quotas, and platform usage guidance to balance experimentation, responsible AI governance, availability, and cost control.
- Translated complex platform behavior and emerging AI capabilities into clear technical guidance for teams adopting enterprise data and analytics services.

Georgia Tech Research Institute — Atlanta, GA June 2024 – January 2025
Graduate Research Assistant — Sensors and Electromagnetic Applications Laboratory

- Conducted human-AI teaming research involving AI digital twins, human-centered design, and human-systems engineering.
- Derived and documented requirements for AI systems involving ethics, safety, situation awareness, user needs, and societal values.
- Evaluated AI systems for performance, usability, reliability, accountability, safety, and user effectiveness across the development lifecycle.

UNC Health — Morrisville, NC June 2024 – October 2024
Data Scientist — Enterprise Data Science Team

- Led Databricks development in Python and SQL for a Length of Stay modeling initiative using linear regression, random forest, and gradient boosting methods.
- Compared candidate models and documented technical findings to support model evaluation, analytic decision-making, and potential operational use.
- Created an interactive SharePoint site and process visualizations to improve access to AI workflow guidance and support knowledge transfer.

Nationwide — Columbus, OH January 2024 – May 2024
Data Engineer Intern — Data Auditing and Analytics

- Gathered requirements from the FinTech Data Ingestion Team and translated them into data curation, validation, and research-ready preparation activities.
- Investigated data sources for completeness and accuracy, developed Python and SQL transformations in Databricks, and tested BI connectivity for business partners.
- Built workflow timelines and delivery estimates, improving visibility into dependencies, handoffs, and expected cycle time.

Georgia Tech Research Institute — Atlanta, GA May 2023 – July 2023
Research Intern — Electro-Optical Systems Laboratory

- Developed and evaluated CNN and TCN models for vehicle-image classification, achieving 98% validation accuracy in distinguishing wheeled versus tracked vehicles.
- Produced readable, reusable Python and Jupyter Notebook documentation to support reproducibility and knowledge transfer.

Intel Corporation — Remote July 2022 – September 2022
Research Intern — AI Healthcare Drive Program

- Developed a CNN image-segmentation approach for brain-ultrasound tumor identification and prepared reusable OneAPI examples for future developer training.

Intel Corporation — Remote July 2021 – September 2021
Ph.D. Researcher — AI Policy Emerging Technology Program

- Conducted structured literature reviews using PRISMA methodology and synthesized emerging-technology research for chief division leaders.
- Composed a survey paper on emerging technology trends and delivered a 20-minute research presentation to executive stakeholders.

TIAA — Charlotte, NC July 2018 – October 2019
Technical Associate — Rotational Program

- Served as a business-technology liaison during migration of an enterprise data lake using Apache Kafka, connecting business needs with technical implementation decisions.
- Developed chatbots, virtual assistants, and NLP-based machine-translation capabilities for a digital retirement platform.
- Served as Scrum Master for small development cycles, coordinating interns and junior developers through delivery and knowledge transfer.

Nationwide — Columbus, OH May 2016 – August 2016

IT Data Scientist Intern — Data Management and Analytics

- Designed data-facsimile frameworks that reduced insurance claim filing time by 15% and built cross-browser JavaScript interfaces to support operational use.

AWARDS/CERTIFICATIONS

Inaugural Trusted Data Impact Spotlight Award — Nationwide HQ, Enterprise 2026 Hackathon, July 2026

Databricks Data Analyst Associate Certification – Data and AI Summit, San Francisco, CA June 2026